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634. MYELOPROLIFERATIVE SYNDROMES: CLINICAL | DECEMBER 07, 2017

Predictors of Vascular Events and Disease Transformation in Chinese Patients with Myeloproliferative Neoplasms during Long-Term Follow-up

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Blood blood **(2017) 130 (Suppl_1) : 5273.**

http://doi.org/10.1182/blood.V130.Suppl_1.5273.5273

Abstract

Introduction and objectives: Myeloproliferative neoplasms (MPN) are a group of heterogeneous clonal hematopoietic disorders that comprise polycythemia vera (PV), essential thrombocythemia (ET) and primary myelofibrosis (PMF). There is an increased risk of hemorrhage, thrombosis, cardiovascular (CV) event and transformation to myelofibrosis or acute leukemia. This study aims to review the clinicopathological characteristics and outcome, and prognostic indicators in a large Chinese population of patients with MPN.

Methods: Patients with the diagnosis of PV, ET, PMF and MPN-unclassifiable were recruited from 2012 to 2016 in a tertiary referral center in Hong Kong. Retrospective data from the date of diagnosis were retrieved from the clinical records and reviewed by independent investigators. Progression-free survival (PFS) was defined as the duration from diagnosis to transformation to myelofibrosis, acute leukemia or death. Overall survival (OS) was defined from the duration from diagnosis to death or last follow-up. Categorical variables were compared using Pearson's chi-square or Fisher's exact test. PFS and OS were evaluated by Kaplan-Meier analysis and were compared by log-rank test. Univariate and multivariate analyses were performed with the Cox proportional hazard model.

Results: 285 patients (142 men, 143 women) were included. PV, ET, PMF and MPN-U accounted for 22%, 60.0%, 10% and 8% of cases respectively (Table 1). Cytogenetic information at diagnosis was

available in 47.7% (n=136) of the patients, of which 83% (N=113) had normal karyotypes. No other recurrent chromosomal abnormalities were observed apart from loss of chromosome Y. Among 230 molecularly annotated patients, 70% (N=162) harbored *JAK2* V617F mutation. In 28 *JAK2* V617F negative patients, 50% (N=14) and 4% (N=1) had *CALR* and *JAK2* exon 12 mutations respectively. 13 patients were triple negative for *JAK2*, *CALR* and *MPL* mutations. Of all patients, treatment included aspirin prophylaxis (92%, N=263), hydroxyurea (92%, N=261), pegylated interferon (10.5%, N=30) and ruxolitinib (6.3%, N=18). CV events were observed in 45 (16%) patients, including 21 transient ischemic attacks, 14 ischemic strokes, 7 acute myocardial infarctions and 2 acute limb ischemia. 34 (76%) events occurred at diagnosis and 11 (24%) occurred while patients were on treatment. Significant risks associated with CV events were age > 50 years (p=0.002) and hypertension (p=0.038). Diabetes mellitus (p=0.163), hyperlipidemia (p=0.699), smoking (p=0.085) and concurrent ischemic heart disease (p=0.470) were not shown to be significant. Progression to MF was observed in 17% (n=11) of PV patients (median follow-up: 14.7, 9.5-27.6 years) and 12% (n=21) of ET patients (median follow-up:10.7, 1.5 - 22.6 years). Leukemic transformation occurred in 9 patients (3%) (ET, N=3, primary MF, N=1, secondary MF, N=5) at a median follow-up of 10.4 (1.0 - 22.6) years. Age >60 years and presence of splenomegaly at diagnosis were associated with inferior PFS; and age >60 years was associated with infereior OS in PV/ET patients (Figure 1). Based on these clinical criteria, PV/ET patients were divided into 3 groups, viz., age > 60 years with splenomegaly (n=8), age < 60 years without splenomegaly (n=83), and others (n=97). The corresponding median PFS were 12.9, 25.5 and 17.4 years; whereas the corresponding median OS were 15.5, 15.2 and 27.2 years. Age>60 years and splenomegaly remained significant risks for inferior PFS in univariate analysis, whereas on multivariate analysis, splenomegaly was only the significant risk for inferior PFS(Figure 2).

Conclusion: This study showed that age >50 years and hypertension were associated with increased risk of cardiovascular events in MPN patients. Age > 60 years and splenomegaly on presentation were significant risks for disease transformation in PV/ET.

Table 1. Clinicopathological features of 285 patients with myeloproliferative neoplasm

	PV (n=63)	ET (n=171)	PMF (n=28)	MPN-U (n=23)
Gender				
Male	33 (52%)	77 (45%)	20 (71%)	12
Female	30 (48%)	94 (55%)	8 (29%)	11
Age (median, range) (yrs)	61 (25-92)	61 (23-89)	65 (23-96)	69 (30-85)
Hb (median, range) (g/dL)	18.1 (15.6-23.1)	13.4 (8.4-16.6)	11.8 (7.3-19.0)	12.1 (7.0-18.4)
Hematocrit (median, range)	0.54 (0.43-0.70)	0.40 (3.0-0.50)	0.36 (0.21-0.56)	0.38 (0.22-0.54)
WCC (median, range) ($x10^9$/L)	15.5 (6.4-46.8)	9.3 (0.8-42.8)	15.0 (1.9-75.8)	11.7 (5.9-35.8)
Platelet (median, range) ($x10^9$/L)	558 (212-1025)	784 (389-2304)	636 (39-1444)	819 (184-1448)
LDH (median, range) (IU/L)	278 (158-1483)	234 (91-990)	366 (157-1346)	220 (154-620)
Cytogenetics at diagnosis				
Normal	14 (22%)	79 (46%)	11 (39%)	9 (39%)
Abnormal	4 (6%)	1 (1%)	5 (18%)	3 14%)
Poor growth	0	5 (3%)	3 (11%)	2 (10%)
Not available	45 (71%)	86 (50%)	9 (32%)	9 (43%)
Gene mutations				
*JAK2*V617F				
Results available	48	134	27	21
Positive	46 (96%)	88 (66%)	17 (63%)	11 (52%)
JAK2 exon 12	1	0	0	0
CALR	0	11	3	0
Type 1	0	5	1	0
Type 2	0	3	2	0
Others	0	3	0	0
MPL	0	0	0	0
Triple negative	1	7	2	3
Splenomegaly at diagnosis	18 (29%)	17 (10%)	14 (50%)	4 (19%)
Comorbidities				
Diabetes mellitus	6 (10%)	26 (15%)	4 (14%)	6 (29%)
Hypertension	12 (19%)	45 (26%)	9 (32%)	6 (29%)
Hyperlipidemia	1 (2%)	10 (6%)	1 (4%)	1 (5%)
History of smoking	6 (10%)	14 (8%)	4 (14%)	1 (5%)
Ischemic heart disease	2 (3%)	7 (4%)	4 (14%)	1 (5%)
Treatment				
Aspirin	62 (98%)	166 (97%)	17 (65%)	18 (86%)
Phlebotomy	10 (16%)	0	1 (4%)	0
Hydroxyurea (HU)	60 (95%)	159 (93%)	22 (79%)	20 (95%)
Anagrelide	1 (2%)	8 (5%)	2 (7%)	0
Pegylated-interferon	7 (11%)	21 (12%)	2 (7%)	0
Ruxolitinib	4 (6%)	7 (4%)	7 (25%)	0

Figure 1. Impact of age and splenomegaly at diagnosis on PFS and OS in PV/ET

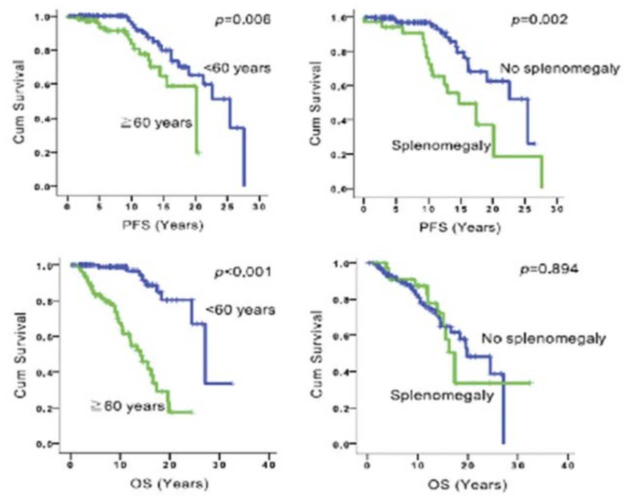
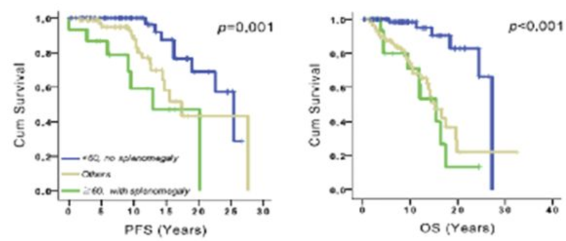


Figure 2. PV/ET risk groups based on age and presence of splenomegaly at diagnosis



Disclosures

No relevant conflicts of interest to declare.

Author notes

*Asterisk with author names denotes non-ASH members.